of the magnetron and the oven cavity become problematic), and at much higher frequencies, the microwave energy cooks the food very unevenly (mostly the surface layer). Whereas microwaves are readily absorbed by water, fats, and sugars, they can penetrate through most ceramics, glass, or plastics without loss of energy, thereby imparting little or no heat to those materials.

► Microwaves are reflected by metal surfaces, so they can bounce around the interior of the chamber, be absorbed by the food, but not escape to the outside. ◄

Oven Operation

To generate high-power microwaves (∼700 watts), the microwave oven uses a ***magnetron tube*** (**Fig. TF3-2**), which requires the application of a voltage on the order of 4000 volts. The typical household voltage of 115 volts is increased to the required voltage level through a ***high-voltage transformer***. The microwave energy generated by the magnetron is transferred into a cooking chamber designed to contain the microwaves within it through the use of metal surfaces and safety Interlock switches.

If the oven door is made of a glass panel, a ***metal screen*** or a layer of conductive mesh is attached to it to ensure the necessary shielding; microwaves cannot pass through the metal screen if the mesh width is much smaller than the wavelength of the microwave ($\lambda \approx 12$ cm at 2.5 GHz). In the chamber, the microwave energy establishes a ***standing-wave pattern***, which leads to an uneven distribution. This is mitigated by using a rotating ***metal stirrer*** that disperses the microwave energy to different parts of the chamber.

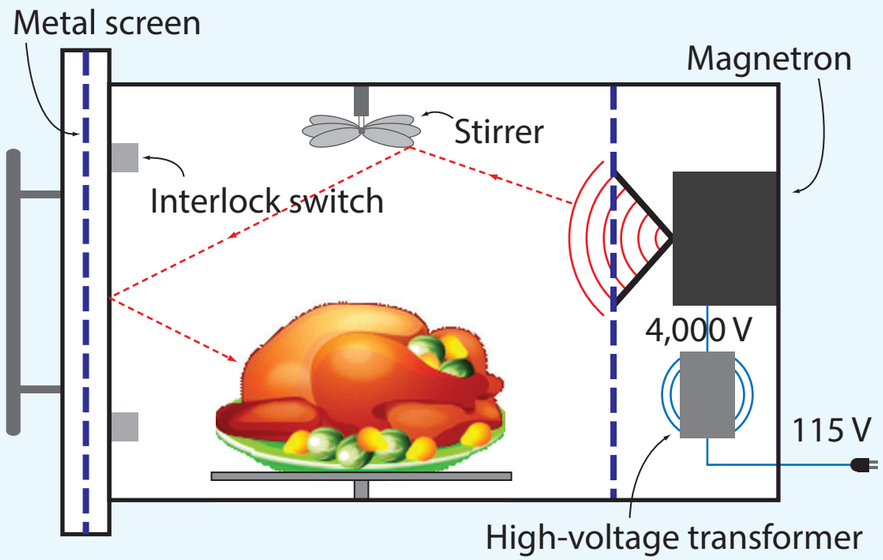


Figure TF3-2 Microwave oven cavity.

磁控管和烤箱腔体的内部结构会出现问题），并且在更高的频率下，微波能量对食物的烹饪非常不均匀（主要是表面层）。虽然微波很容易被水、脂肪和糖吸收，但它们可以穿透大多数陶瓷、玻璃或塑料而不会损失能量，从而向这些材料传递很少或不传递热量。

► 微波会被金属表面反射，因此它们可以在室内部反弹，被食物吸收，但不会逸出到外部。 ◄

烤箱操作

为了产生高功率微波（∼ 700瓦），微波炉使用磁控管（图TF3-2），需要施加约4000伏的电压。典型的家庭电压为 115 伏，通过高压变压器增加到所需的电压水平。磁控管产生的微波能量被转移到烹饪室中，该烹饪室设计为通过使用金属表面和安全联锁开关将微波容纳在其中。

如果烤箱门由玻璃面板制成，则在其上附有金属丝网或一层导电网，以确保必要的屏蔽；如果网孔宽度远小于微波的波长（2.5 GHz 时$\lambda \approx 12$ cm），微波就无法穿过金属筛网。在腔室中，微波能量形成驻波模式，从而导致分布不均匀。通过使用旋转金属搅拌器将微波能量分散到腔室的不同部分可以减轻这种情况。

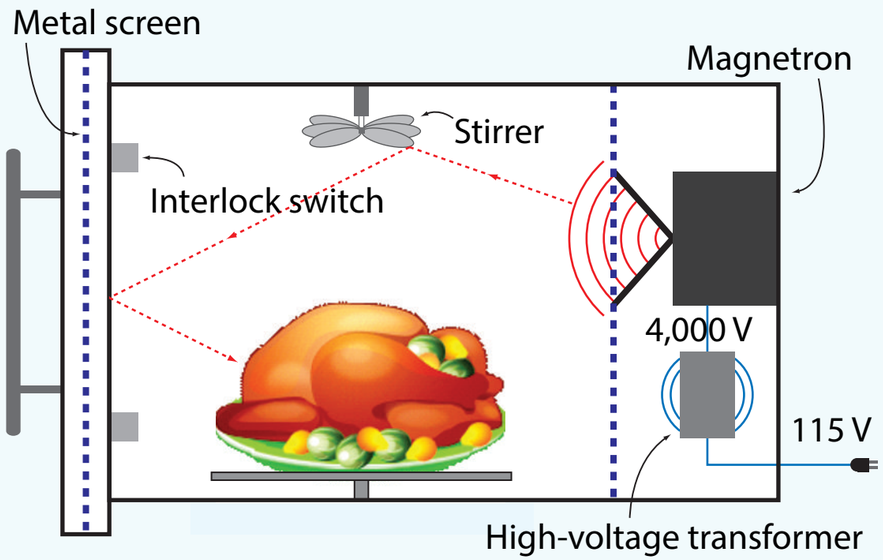


图 TF3-2 微波炉腔体。

2-9 Power Flow on a Lossless Transmission Line

Our discussion thus far has focused on the voltage and current attributes of waves propagating on a transmission line. Now we examine the flow of power carried by the incident and reflected waves. We begin by reintroducing Eqs. (2.63a) and (2.63b) with $z = -d$:

$$\tilde{V}(d) = V_0^+(e^{j\beta d} + \Gamma e^{-j\beta d}), \tag{2.98a}$$

$$\tilde{I}(d) = \frac{V_0^+}{Z_0}(e^{j\beta d} - \Gamma e^{-j\beta d}). \tag{2.98b}$$

In these expressions, the first terms represent the incident-wave voltage and current, and the terms involving Γ represent the reflected-wave voltage and current. The time-domain expressions for the voltage and current at location d from the load are obtained by transforming Eq. (2.98) to the time domain:

$$\begin{aligned} v(d,t) &= \mathfrak{Re}[\tilde{V}e^{j\omega t}] \\ &= \mathfrak{Re}[|V_0^+|e^{j\phi^+}(e^{j\beta d} + |\Gamma|e^{j\theta_r}e^{-j\beta d})e^{j\omega t}] \\ &= |V_0^+|[\cos(\omega t + \beta d + \phi^+) \\ &\quad + |\Gamma|\cos(\omega t - \beta d + \phi^+ + \theta_r)], \end{aligned} \tag{2.99a}$$

$$\begin{aligned} i(d,t) &= \frac{|V_0^+|}{Z_0}[\cos(\omega t + \beta d + \phi^+) \\ &\quad - |\Gamma|\cos(\omega t - \beta d + \phi^+ + \theta_r)], \end{aligned} \tag{2.99b}$$

where we used the relations $V_0^+ = |V_0^+|e^{j\phi^+}$ and $\Gamma = |\Gamma|e^{j\theta_r}$ that were introduced earlier as Eqs. (2.31a) and (2.62), respectively.

2-9.1 Instantaneous Power

The instantaneous power carried by the transmission line is equal to the product of $v(d,t)$ and $i(d,t)$:

$$\begin{aligned} P(d,t) &= v(d,t)\, i(d,t) \\ &= |V_0^+|[\cos(\omega t + \beta d + \phi^+) \\ &\quad + |\Gamma|\cos(\omega t - \beta d + \phi^+ + \theta_r)] \\ &\quad \times \frac{|V_0^+|}{Z_0}[\cos(\omega t + \beta d + \phi^+) \\ &\quad - |\Gamma|\cos(\omega t - \beta d + \phi^+ + \theta_r)] \\ &= \frac{|V_0^+|^2}{Z_0}[\cos^2(\omega t + \beta d + \phi^+) \\ &\quad - |\Gamma|^2\cos^2(\omega t - \beta d + \phi^+ + \theta_r)] \quad \text{(W)}. \end{aligned} \tag{2.100}$$

Per our earlier discussion in connection with Eq. (1.31), if the signs preceding ωt and βd in the argument of the cosine term are both positive or both negative, then the cosine term represents a wave traveling in the negative d direction. Since d points from the load to the generator, the first term in Eq. (2.100) represents the ***instantaneous incident power*** traveling towards the load. This is the power that would be delivered to the load in the absence of wave reflection (when $\Gamma = 0$). Because βd is preceded by a minus sign in the argument of the cosine of the second term in Eq. (2.100), that term represents the ***instantaneous reflected power*** traveling in the $+d$ direction—away from the load. Accordingly, we label these two power components

$$P^i(d,t) = \frac{|V_0^+|^2}{Z_0}\cos^2(\omega t + \beta d + \phi^+) \quad \text{(W)}, \tag{2.101a}$$

$$P^r(d,t) = -|\Gamma|^2\frac{|V_0^+|^2}{Z_0}\cos^2(\omega t - \beta d + \phi^+ + \theta_r) \quad \text{(W)}. \tag{2.101b}$$

Using the trigonometric identity

$$\cos^2 x = \tfrac{1}{2}(1 + \cos 2x),$$

the expressions in Eq. (2.101) can be rewritten as

$$P^i(d,t) = \frac{|V_0^+|^2}{2Z_0}[1 + \cos(2\omega t + 2\beta d + 2\phi^+)], \tag{2.102a}$$

$$P^r(d,t) = -|\Gamma|^2\frac{|V_0^+|^2}{2Z_0}[1 + \cos(2\omega t - 2\beta d + 2\phi^+ + 2\theta_r)]. \tag{2.102b}$$

We note that in each case, the instantaneous power consists of a dc term (not varying in time) and an ac term that oscillates at an angular frequency of 2ω.

► The power oscillates at twice the rate of the voltage or current. ◄

2-9.2 Time-Average Power

From a practical standpoint, we usually are more interested in the time-average power flowing along the transmission line, $P_{av}(d)$, than in the instantaneous power $P(d,t)$. To compute $P_{av}(d)$, we can use a time-domain approach or a computationally simpler phasor-domain approach. For completeness, we consider both.

2-9 无损传输线上的功率流

到目前为止，我们的讨论主要集中在传输线上传播的波的电压和电流属性。现在我们研究入射波和反射波所携带的能量流。我们首先重新引入方程。 (2.63a) 和 (2.63b) 与 z = −d：

$$\tilde{V}(d) = V_0^+(e^{j\beta d} + \Gamma e^{-j\beta d}), \tag{2.98a}$$

$$\tilde{I}(d) = \frac{V_0^+}{Z_0}(e^{j\beta d} - \Gamma e^{-j\beta d}). \tag{2.98b}$$

在这些表达式中，第一项代表入射波电压和电流，涉及Γ的项代表反射波电压和电流。通过对式（1）进行变换，得到负载 d 处电压和电流的时域表达式。 (2.98) 到时域：

$$\begin{aligned} v(d,t) &= \mathfrak{Re}[\tilde{V}e^{j\omega t}] \\ &= \mathfrak{Re}[|V_0^+|e^{j\phi^+}(e^{j\beta d} + |\Gamma|e^{j\theta_r}e^{-j\beta d})e^{j\omega t}] \\ &= |V_0^+|[\cos(\omega t + \beta d + \phi^+) \\ &\quad + |\Gamma|\cos(\omega t - \beta d + \phi^+ + \theta_r)], \end{aligned} \tag{2.99a}$$

$$\begin{aligned} i(d,t) &= \frac{|V_0^+|}{Z_0}[\cos(\omega t + \beta d + \phi^+) \\ &\quad - |\Gamma|\cos(\omega t - \beta d + \phi^+ + \theta_r)], \end{aligned} \tag{2.99b}$$

其中我们使用了之前作为等式引入的关系 $V_0^+ = |V_0^+|e^{j\phi^+}$ 和 $\Gamma = |\Gamma|e^{j\theta_r}$。分别为（2.31a）和（2.62）。

2-9.1 瞬时功率

传输线承载的瞬时功率等于 v(d,t) 和 i(d,t) 的乘积：

$$\begin{aligned} P(d,t) &= v(d,t)\, i(d,t) \\ &= |V_0^+|[\cos(\omega t + \beta d + \phi^+) \\ &\quad + |\Gamma|\cos(\omega t - \beta d + \phi^+ + \theta_r)] \\ &\quad \times \frac{|V_0^+|}{Z_0}[\cos(\omega t + \beta d + \phi^+) \\ &\quad - |\Gamma|\cos(\omega t - \beta d + \phi^+ + \theta_r)] \\ &= \frac{|V_0^+|^2}{Z_0}[\cos^2(\omega t + \beta d + \phi^+) \\ &\quad - |\Gamma|^2\cos^2(\omega t - \beta d + \phi^+ + \theta_r)] \quad \text{(W)}. \end{aligned} \tag{2.100}$$

根据我们之前关于方程式的讨论。 （1.31），如果余弦项的自变量中ω t和β d之前的符号均为正或均为负，则余弦项表示沿负d方向传播的波。由于 d 从负载指向发电机，因此方程中的第一项。 (2.100) 表示流向负载的瞬时入射功率。这是在没有波反射的情况下（当 Γ = 0 时）传递到负载的功率。因为在方程第二项的余弦参数中，β d 前面有一个负号。 （2.100），该术语表示沿 +d 方向（远离负载）传播的瞬时反射功率。因此，我们将这两个功率组件标记为

$$P^i(d,t) = \frac{|V_0^+|^2}{Z_0}\cos^2(\omega t + \beta d + \phi^+) \quad \text{(W)}, \tag{2.101a}$$

$$P^r(d,t) = -|\Gamma|^2\frac{|V_0^+|^2}{Z_0}\cos^2(\omega t - \beta d + \phi^+ + \theta_r) \quad \text{(W)}. \tag{2.101b}$$

使用三角恒等式

$$\cos^2 x = \tfrac{1}{2}(1 + \cos 2x),$$

方程中的表达式(2.101) 可以重写为

$$P^i(d,t) = \frac{|V_0^+|^2}{2Z_0}[1 + \cos(2\omega t + 2\beta d + 2\phi^+)], \tag{2.102a}$$

$$P^r(d,t) = -|\Gamma|^2\frac{|V_0^+|^2}{2Z_0}[1 + \cos(2\omega t - 2\beta d + 2\phi^+ + 2\theta_r)]. \tag{2.102b}$$

我们注意到，在每种情况下，瞬时功率由直流项（不随时间变化）和以 2ω 角频率振荡的交流项组成。

► 功率以电压或电流速率的两倍振荡。 ◄

2-9.2 时间平均功率

从实际角度来看，我们通常对沿传输线流动的时间平均功率 $P_{av}(d)$ 更感兴趣，而不是瞬时功率 P(d,t)。为了计算 $P_{av}(d)$，我们可以使用时域方法或计算更简单的相量域方法。为了完整起见，我们同时考虑两者。

Time-Domain Approach

The time-average power is equal to the instantaneous power averaged over one time period $T = 1/f = 2\pi/\omega$. For the incident wave, its time-average power is

$$P^{i}_{av}(d) = \frac{1}{T}\int_0^T P^{i}(d,t)\,dt = \frac{\omega}{2\pi}\int_0^{2\pi/\omega} P^{i}(d,t)\,dt. \quad (2.103)$$

Upon inserting Eq. (2.102a) into Eq. (2.103) and performing the integration, we obtain

$$P^{i}_{av} = \frac{|V_0^+|^2}{2Z_0} \quad \text{(W)}, \quad (2.104)$$

which is identical with the dc term of $P^{i}(d,t)$ given by Eq. (2.102a). A similar treatment for the reflected wave gives

$$P^{r}_{av} = -|\Gamma|^2\frac{|V_0^+|^2}{2Z_0} = -|\Gamma|^2 P^{i}_{av}. \quad (2.105)$$

► The average reflected power is equal to the average incident power, diminished by a multiplicative factor of $|\Gamma|^2$. ◄

Note that the expressions for P^{i}_{av} and P^{r}_{av} are independent of d, which means that the time-average powers carried by the incident and reflected waves do not change as they travel along the transmission line. This is as expected, because the transmission line is lossless.

The ***net average power flowing towards (and then absorbed by) the load*** shown in **Fig. 2-23** is

$$P_{av} = P^{i}_{av} + P^{r}_{av} = \frac{|V_0^+|^2}{2Z_0}[1 - |\Gamma|^2] \quad \text{(W)}. \quad (2.106)$$

Phasor-Domain Approach

For any propagating wave with voltage and current phasors $\widetilde{V}$ and $\widetilde{I}$, a useful formula for computing the time-average power is

$$P_{av} = \tfrac{1}{2}\Re\mathrm{e}\left[\widetilde{V}\cdot\widetilde{I}^*\right], \quad (2.107)$$

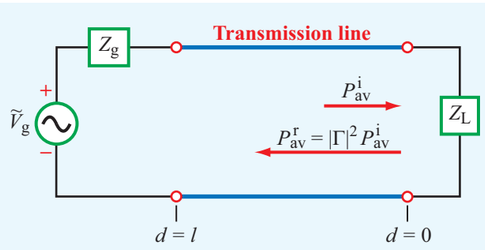


Figure 2-23 The time-average power reflected by a load connected to a lossless transmission line is equal to the incident power multiplied by $|\Gamma|^2$.

where $\widetilde{I}^*$ is the complex conjugate of $\widetilde{I}$. Application of this formula to Eqs. (2.98a) and (2.98b) gives

$$\begin{aligned} P_{av} &= \frac{1}{2}\Re\mathrm{e}\left[V_0^+(e^{j\beta d} + \Gamma e^{-j\beta d})\frac{V_0^{+*}}{Z_0}(e^{-j\beta d} - \Gamma^* e^{j\beta d})\right] \\ &= \frac{1}{2}\Re\mathrm{e}\left[\frac{|V_0^+|^2}{Z_0}(1 - |\Gamma|^2 + \Gamma e^{-j2\beta d} - \Gamma^* e^{j2\beta d})\right] \\ &= \frac{|V_0^+|^2}{2Z_0}\left\{[1 - |\Gamma|^2] + \Re\mathrm{e}\left[|\Gamma|e^{-j(2\beta d - \theta_r)} - |\Gamma|e^{j(2\beta d - \theta_r)}\right]\right\} \\ &= \frac{|V_0^+|^2}{2Z_0}\left\{[1 - |\Gamma|^2] + |\Gamma|[\cos(2\beta d - \theta_r) - \cos(2\beta d - \theta_r)]\right\} \\ &= \frac{|V_0^+|^2}{2Z_0}[1 - |\Gamma|^2], \end{aligned} \quad (2.108)$$

which is identical to Eq. (2.106).

Exercise 2-17: For a 50 Ω lossless transmission line terminated in a load impedance $Z_L = (100 + j50)$ Ω, determine the fraction of the average incident power reflected by the load.

Answer: 20%. (See EM.)

Exercise 2-18: For the line of Exercise 2-17, what is the magnitude of the average reflected power if $|V_0^+| = 1$ V?

Answer: $P^{r}_{av} = 2$ (mW). (See EM.)

Concept Question 2-15: According to Eq. (2.102b), the instantaneous value of the reflected power depends on the phase of the reflection coefficient θ_r, but the average reflected power given by Eq. (2.105) does not. Explain.

时域方法

时间平均功率等于一个时间段T = 1/ f = 2π/ω 内平均的瞬时功率。对于入射波，其时均功率为

$$P^{i}_{av}(d) = \frac{1}{T}\int_0^T P^{i}(d,t)\,dt = \frac{\omega}{2\pi}\int_0^{2\pi/\omega} P^{i}(d,t)\,dt. \quad (2.103)$$

插入方程后。 (2.102a) 代入方程。 (2.103) 并进行积分，我们得到

$$P^{i}_{av} = \frac{|V_0^+|^2}{2Z_0} \quad \text{(W)}, \quad (2.104)$$

这与式 (1) 给出的 $P^{i}(d,t)$ 的直流项相同。 (2.102a) 。对反射波的类似处理给出

$$P^{r}_{av} = -|\Gamma|^2\frac{|V_0^+|^2}{2Z_0} = -|\Gamma|^2 P^{i}_{av}. \quad (2.105)$$

► 平均反射功率等于平均入射功率减去乘法因子$|\Gamma|^2$。◄

请注意，P^{i}_{av} 和 P^{r}_{av} 的表达式与 d 无关，这意味着入射波和反射波所携带的时间平均功率在沿传输线传播时不会改变。这是符合预期的，因为传输线是无损的。

流向负载（然后被负载吸收）的净平均功率如图 2-23 所示：

$$P_{av} = P^{i}_{av} + P^{r}_{av} = \frac{|V_0^+|^2}{2Z_0}[1 - |\Gamma|^2] \quad \text{(W)}. \quad (2.106)$$

相量域方法

对于任何具有电压和电流相量 $\widetilde{}$V 和 $\widetilde{}$I 的传播波，计算时间平均功率的有用公式为

$$P_{av} = \tfrac{1}{2}\Re\mathrm{e}\left[\widetilde{V}\cdot\widetilde{I}^*\right], \quad (2.107)$$

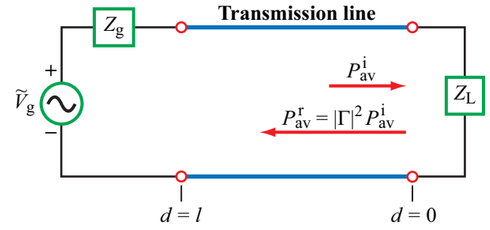


图2-23 连接到无损传输线的负载反射的时间平均功率等于入射功率乘以$|\Gamma|^2$。

其中 $\widetilde{}$I* 是 $\widetilde{}$I 的复共轭。将此公式应用于方程式。 (2.98a) 和 (2.98b) 给出

$$\begin{aligned} P_{av} &= \frac{1}{2}\Re\mathrm{e}\left[V_0^+(e^{j\beta d} + \Gamma e^{-j\beta d})\frac{V_0^{+*}}{Z_0}(e^{-j\beta d} - \Gamma^* e^{j\beta d})\right] \\ &= \frac{1}{2}\Re\mathrm{e}\left[\frac{|V_0^+|^2}{Z_0}(1 - |\Gamma|^2 + \Gamma e^{-j2\beta d} - \Gamma^* e^{j2\beta d})\right] \\ &= \frac{|V_0^+|^2}{2Z_0}\left\{[1 - |\Gamma|^2] + \Re\mathrm{e}\left[|\Gamma|e^{-j(2\beta d - \theta_r)} - |\Gamma|e^{j(2\beta d - \theta_r)}\right]\right\} \\ &= \frac{|V_0^+|^2}{2Z_0}\left\{[1 - |\Gamma|^2] + |\Gamma|[\cos(2\beta d - \theta_r) - \cos(2\beta d - \theta_r)]\right\} \\ &= \frac{|V_0^+|^2}{2Z_0}[1 - |\Gamma|^2], \end{aligned} \quad (2.108)$$

这与等式相同。 (2.106)。

练习 2-17：对于终止于负载阻抗 $Z_L = (100 + j50)$ Ω 的 50 Ω 无损传输线，确定负载反射的平均入射功率的分数。

答案：20%。 （参见EM。）

练习 2-18：对于练习 2-17 的直线，如果 $|V_0^+| = 1$ V，平均反射功率的大小是多少？

Answer: $P^{r}_{av} = 2$ (mW). (See EM.)

概念问题 2-15：根据方程式(2.102b)中，反射功率的瞬时值取决于反射系数θ_r的相位，但平均反射功率由式(2.102b)给出。 (2.105) 则不然。解释。

Concept Question 2-16: What is the average power delivered by a lossless transmission line to a reactive load?

Concept Question 2-17: What fraction of the incident power is delivered to a matched load?

Concept Question 2-18: Verify that

$$\frac{1}{T}\int_0^T \cos^2\left(\frac{2\pi t}{T}+\beta d+\phi\right)dt=\frac{1}{2},$$

regardless of the values of d and ϕ, so long as neither is a function of t.

2-10 The Smith Chart

The ***Smith chart***, developed by P. H. Smith in 1939, is a widely used graphical tool for analyzing and designing transmission-line circuits. Even though it was originally intended to facilitate calculations involving complex impedances, the Smith chart has become an important avenue for comparing and characterizing the performance of microwave circuits. As the material in this and the next section demonstrates, use of the Smith chart not only avoids tedious manipulations of complex numbers, but it also allows an engineer to design impedance-matching circuits with relative ease.

2-10.1 Parametric Equations

The reflection coefficient Γ is, in general, a complex quantity composed of a magnitude $|\Gamma|$ and a phase angle θ_r or, equivalently, a real part Γ_r and an imaginary part Γ_i,

$$\Gamma=|\Gamma|e^{j\theta_r}=\Gamma_r+j\Gamma_i, \tag{2.109}$$

where

$$\Gamma_r=|\Gamma|\cos\theta_r, \tag{2.110a}$$
$$\Gamma_i=|\Gamma|\sin\theta_r. \tag{2.110b}$$

The Smith chart lies in the complex Γ plane. In **Fig. 2-24**, point A represents a reflection coefficient $\Gamma_A=0.3+j0.4$ or, equivalently,

$$|\Gamma_A|=[(0.3)^2+(0.4)^2]^{1/2}=0.5$$

and

$$\theta_{r_A}=\tan^{-1}(0.4/0.3)=53^\circ.$$

Similarly, point B represents $\Gamma_B=-0.5-j0.2$, or $|\Gamma_B|=0.54$ and $\theta_{r_B}=202^\circ$ or, equivalently,

$$\theta_{r_B}=(360^\circ-202^\circ)=-158^\circ.$$

▶ When both Γ_r and Γ_i are negative, θ_r is in the third quadrant in the Γ_r–Γ_i plane. Thus, when using $\theta_r=\tan^{-1}(\Gamma_i/\Gamma_r)$ to compute θ_r, it may be necessary to add or subtract 180° to obtain the correct value of θ_r. ◀

The ***unit circle*** shown in **Fig. 2-24** corresponds to $|\Gamma|=1$. Because $|\Gamma|\le 1$ for a transmission line terminated with a passive load, only that part of the Γ_r–Γ_i plane that lies within the unit circle is useful to us; hence, future drawings will be limited to the domain contained within the unit circle.

Impedances on a Smith chart are represented by their values normalized to the line's characteristic impedance, Z_0. From

$$\Gamma=\frac{Z_L/Z_0-1}{Z_L/Z_0+1}=\frac{z_L-1}{z_L+1}, \tag{2.111}$$

the inverse relation is

$$z_L=\frac{1+\Gamma}{1-\Gamma}. \tag{2.112}$$

The normalized load impedance z_L is, in general, a complex quantity composed of a ***normalized load resistance*** r_L and a ***normalized load reactance*** x_L:

$$z_L=r_L+jx_L. \tag{2.113}$$

Using Eqs. (2.109) and (2.113) in Eq. (2.112), we have

$$r_L+jx_L=\frac{(1+\Gamma_r)+j\Gamma_i}{(1-\Gamma_r)-j\Gamma_i}, \tag{2.114}$$

which can be manipulated to obtain explicit expressions for each of r_L and x_L in terms of Γ_r and Γ_i. This is accomplished by multiplying the numerator and denominator of the right-hand side of Eq. (2.114) by the complex conjugate of the denominator and then separating the result into real and imaginary parts. These steps lead to

$$r_L=\frac{1-\Gamma_r^2-\Gamma_i^2}{(1-\Gamma_r)^2+\Gamma_i^2}, \tag{2.115a}$$
$$x_L=\frac{2\Gamma_i}{(1-\Gamma_r)^2+\Gamma_i^2}. \tag{2.115b}$$

概念问题 2-16：无损传输线向无功负载传送的平均功率是多少？

概念问题 2-17： 入射功率的哪一部分传输至匹配负载？

概念问题 2-18：验证

$$\frac{1}{T}\int_0^T \cos^2\left(\frac{2\pi t}{T}+\beta d+\phi\right)dt=\frac{1}{2},$$

无论 d 和 ϕ 的值如何，只要两者都不是 t 的函数。

2-10 史密斯圆图

史密斯圆图由 P. H. Smith 于 1939 年开发，是一种广泛使用的用于分析和设计传输线电路的图形工具。尽管史密斯圆图最初的目的是促进涉及复数阻抗的计算，但它已成为比较和表征微波电路性能的重要途径。正如本节和下一节中的材料所示，史密斯圆图的使用不仅避免了对复数的繁琐操作，而且还允许工程师相对轻松地设计阻抗匹配电路。

2-10.1 参数方程

一般来说，反射系数 Γ 是一个由幅度 |Γ| 和相位角 θ_r 组成的复数，或者等效地，由实部 Γ_r 和虚部 Γ_i 组成，

$$\Gamma=|\Gamma|e^{j\theta_r}=\Gamma_r+j\Gamma_i, \tag{2.109}$$

在哪里

$$\Gamma_r=|\Gamma|\cos\theta_r, \tag{2.110a}$$
$$\Gamma_i=|\Gamma|\sin\theta_r. \tag{2.110b}$$

史密斯圆图位于复数 Γ 平面中。在图 2-24 中，点 A 代表反射系数 $\Gamma_A=0.3+j0.4$ 或等效地，

$$|\Gamma_A|=[(0.3)^2+(0.4)^2]^{1/2}=0.5$$

和

$$\theta_{r_A}=\tan^{-1}(0.4/0.3)=53^\circ.$$

类似地，点 B 表示 $\Gamma_B=-0.5-j0.2$，或 $|\Gamma_B|=0.54$ 和 $\theta_{r_B}=202^\circ$，或者等效地，

$$\theta_{r_B}=(360^\circ-202^\circ)=-158^\circ.$$

► 当Γ_r和Γ_i均为负数时，θ_r位于Γ_r-Γ_i平面的第三象限。因此，当使用 $\theta_r=\tan^{-1}(\Gamma_i/\Gamma_r)$ 计算 θ_r 时，可能需要加上或减去 180° 才能获得正确的 θ_r 值。 ◀

图2-24所示的单位圆对应于$|\Gamma|=1$。因为$|\Gamma|\le 1$对于以无源负载端接的传输线来说，只有位于单位圆内的Γ_r-Γ_i平面的部分对我们有用；因此，未来的绘图将仅限于单位圆内包含的域。

史密斯圆图上的阻抗由其归一化为线路特征阻抗 Z_0 的值表示。从

$$\Gamma=\frac{Z_L/Z_0-1}{Z_L/Z_0+1}=\frac{z_L-1}{z_L+1}, \tag{2.111}$$

反比关系是

$$z_L=\frac{1+\Gamma}{1-\Gamma}. \tag{2.112}$$

归一化负载阻抗 z_L 通常是由归一化负载电阻 r_L 和归一化负载电抗 x_L 组成的复数：

$$z_L=r_L+jx_L. \tag{2.113}$$

使用方程式。式（2.109）和（2.113） （2.112），我们有

$$r_L+jx_L=\frac{(1+\Gamma_r)+j\Gamma_i}{(1-\Gamma_r)-j\Gamma_i}, \tag{2.114}$$

可以对其进行操作以获得 r_L 和 x_L 中每个以 Γ_r 和 Γ_i 表示的显式表达式。这是通过将等式右侧的分子和分母相乘来实现的。 （2.114）通过分母的复共轭，然后将结果分为实部和虚部。这些步骤导致

$$r_L=\frac{1-\Gamma_r^2-\Gamma_i^2}{(1-\Gamma_r)^2+\Gamma_i^2}, \tag{2.115a}$$
$$x_L=\frac{2\Gamma_i}{(1-\Gamma_r)^2+\Gamma_i^2}. \tag{2.115b}$$

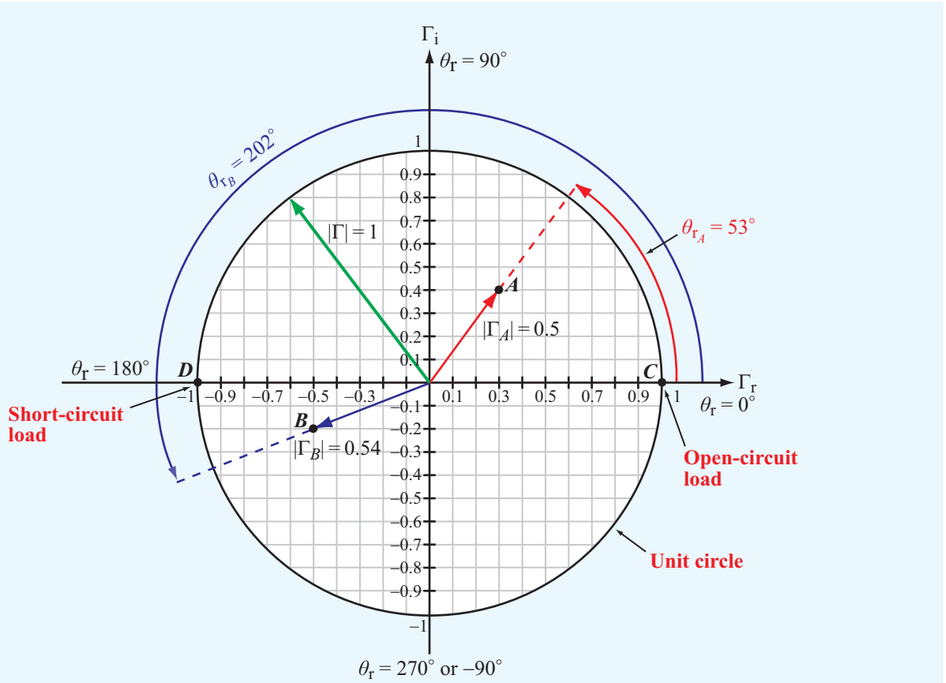


Figure 2-24 The complex Γ plane. Point *A* is at $\Gamma_A = 0.3 + j0.4 = 0.5e^{j53^\circ}$, and point *B* is at $\Gamma_B = -0.5 - j0.2 = 0.54e^{j202^\circ}$. The unit circle corresponds to $|\Gamma| = 1$. At point *C*, $\Gamma = 1$, corresponding to an open-circuit load, and at point *D*, $\Gamma = -1$, corresponding to a short circuit.

Equation (2.115a) implies that there exist many combinations of values for Γ_r and Γ_i that yield the same value for the normalized load resistance r_L. For example, $(\Gamma_r, \Gamma_i) = (0.33, 0)$ gives $r_L = 2$, as does $(\Gamma_r, \Gamma_i) = (0.5, 0.29)$, as well as an infinite number of other combinations. In fact, if we were to plot in the Γ_r–Γ_i plane all possible combinations of Γ_r and Γ_i corresponding to $r_L = 2$, we would obtain the circle labeled $r_L = 2$ in **Fig. 2-25**. Similar circles can be obtained for other values of r_L. After some algebraic manipulations, Eq. (2.115a) can be rearranged into the ***parametric equation*** for the circle in the Γ_r–Γ_i plane corresponding to a given value of r_L:

$$\left(\Gamma_r - \frac{r_L}{1+r_L}\right)^2 + \Gamma_i^2 = \left(\frac{1}{1+r_L}\right)^2. \tag{2.116}$$

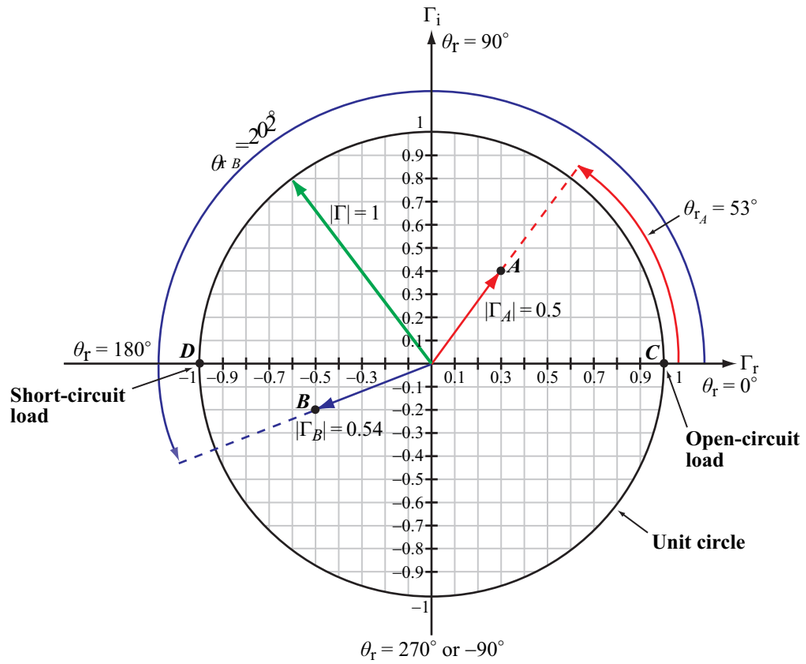
The standard equation for a circle in the *x*–*y* plane with center at (x_0, y_0) and radius *a* is

$$(x - x_0)^2 + (y - y_0)^2 = a^2. \tag{2.117}$$

Comparison of Eq. (2.116) with Eq. (2.117) shows that the r_L circle is centered at $\Gamma_r = r_L/(1 + r_L)$ and $\Gamma_i = 0$, and its radius is $1/(1 + r_L)$. It therefore follows that all r_L-circles pass through the point $(\Gamma_r, \Gamma_i) = (1, 0)$. The largest circle shown in **Fig. 2-25** is for $r_L = 0$, which also is the unit circle corresponding to $|\Gamma| = 1$. This is to be expected, because when $r_L = 0$, $|\Gamma| = 1$ regardless of the magnitude of x_L.

A similar manipulation of the expression for x_L given by Eq. (2.115b) leads to

$$(\Gamma_r - 1)^2 + \left(\Gamma_i - \frac{1}{x_L}\right)^2 = \left(\frac{1}{x_L}\right)^2, \tag{2.118}$$



非图 2-24 复数 Γ 平面。 A 点位于 $\Gamma_A = 0.3 + j0.4 = 0.5e^{j53^\circ}$ 处，B 点位于 $\Gamma_B = -0.5 - j0.2 = 0.54e^{j202^\circ}$ 处。单位循环对应 $|\Gamma| = 1$。在 C 点，$\Gamma = 1$，对应开路负载，在 D 点，$\Gamma = -1$，对应短路循环特。

方程 (2.115a) 意味着存在许多 Γ_r 和 Γ_i 值的组合，它们产生相同的归一化负载电阻 r_L 值。例如，$(\Gamma_r, \Gamma_i) = (0.33, 0)$ 给出 $r_L = 2$，$(\Gamma_r, \Gamma_i) = (0.5, 0.29)$ 以及无数其他组合也是如此。事实上，如果我们在 Γ_r–Γ_i 平面上绘制对应于 $r_L = 2$ 的 Γ_r 和 Γ_i 的所有可能组合，我们将得到图 2-25 中标记为 $r_L = 2$ 的圆。对于 r_L 的其他值也可以获得类似的圆。经过一些代数运算后，方程。 (2.115a) 可以重新整理成对应于给定 r_L 值的 Γ_r–Γ_i 平面中圆的参数方程：

$$\left(\Gamma_r - \frac{r_L}{1+r_L}\right)^2 + \Gamma_i^2 = \left(\frac{1}{1+r_L}\right)^2. \tag{2.116}$$

x-y 平面上以 (x_0, y_0) 为圆心、半径为 a 的圆的标准方程为

$$(x - x_0)^2 + (y - y_0)^2 = a^2. \tag{2.117}$$

等式的比较(2.116) 与等式。 (2.117) 表明 r_L 圆以 $\Gamma_r = r_L/(1 + r_L)$ 和 $\Gamma_i = 0$ 为圆心，半径为 $1/(1 + r_L)$。因此，所有 r_L 圆都经过点 $(\Gamma_r, \Gamma_i) = (1, 0)$。图2-25所示的最大圆是$r_L = 0$时的，它也是对应于$|\Gamma| = 1$的单位圆。这是可以预料的，因为当$r_L = 0$时，无论$x_L$的大小如何，$|\Gamma|$ =都是1。

对等式给出的 x_L 表达式进行类似的操作。 (2.115b) 导致

$$(\Gamma_r - 1)^2 + \left(\Gamma_i - \frac{1}{x_L}\right)^2 = \left(\frac{1}{x_L}\right)^2, \tag{2.118}$$